

Analog Transmission

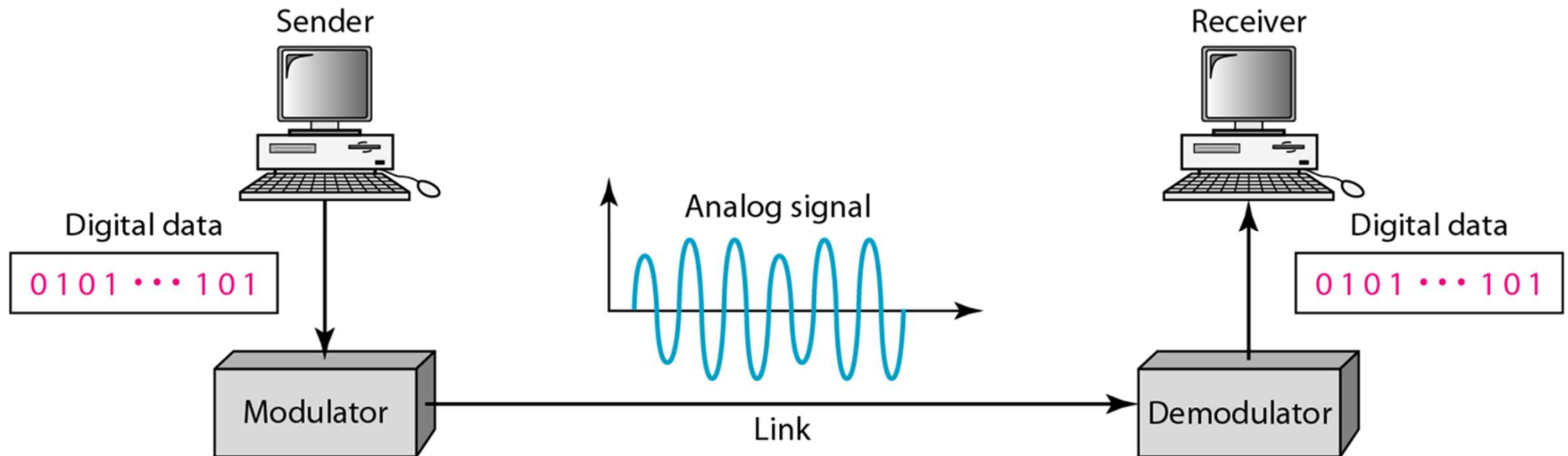
Outline

- Digital-to-Analog Conversion
- Analog-to-Analog Conversion

Digital-to-Analog Conversion

Data Communication Model

- Digital → Analog – (send) – Analog → Digital

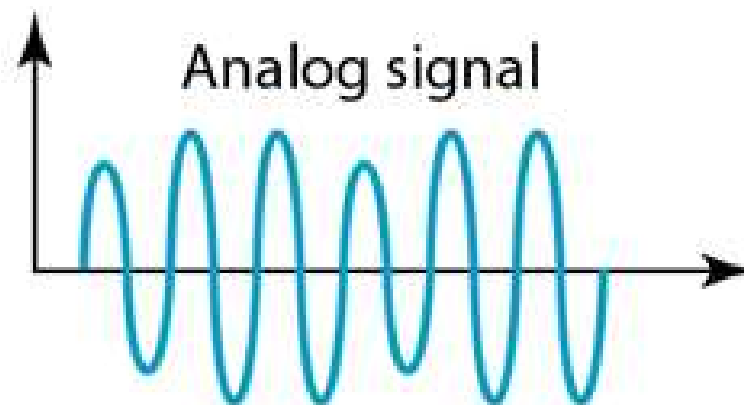


Reason for Converting to Analog Signal

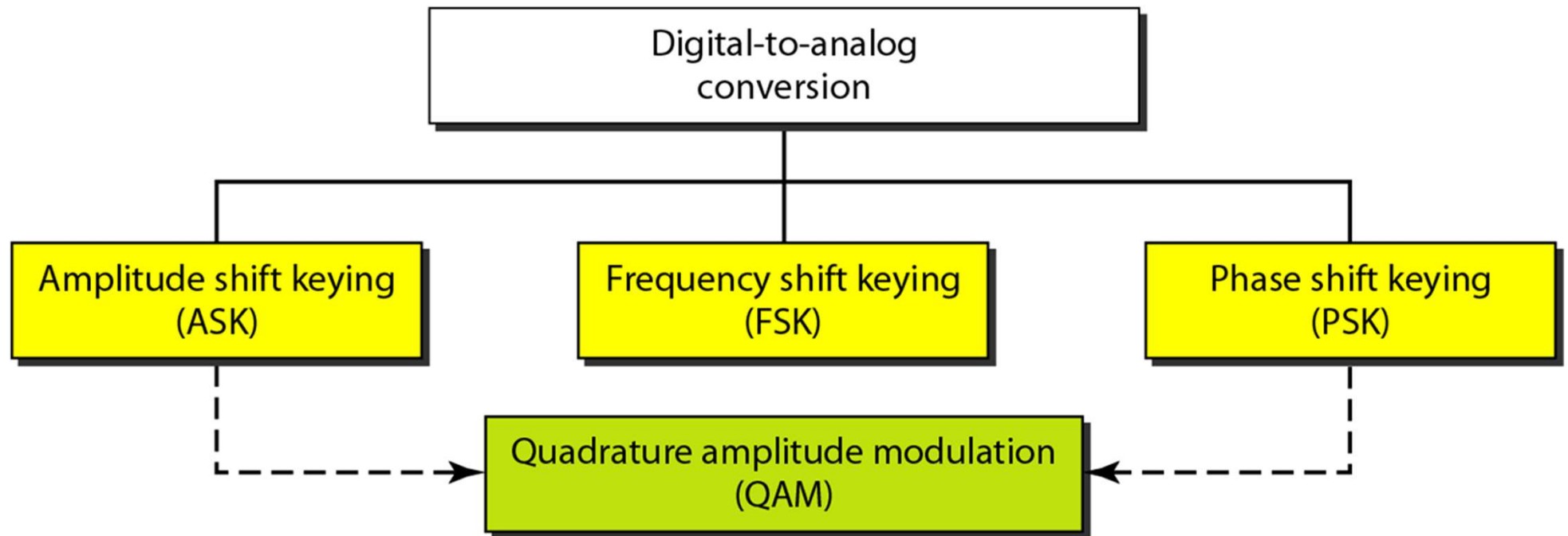
- When the available channel is bandpass (e.g., wireless, shared media), digital data must be converted to analog signals
- Broadcasting services such as TV and radio allocate different frequency bands to different channels
- Different frequency bands are allocated to cell phone users for simultaneous calls

Analog Signal

- Amplitude
 - Frequency
 - Phase
-
- Digital data can be "encoded" using these 3 characteristics

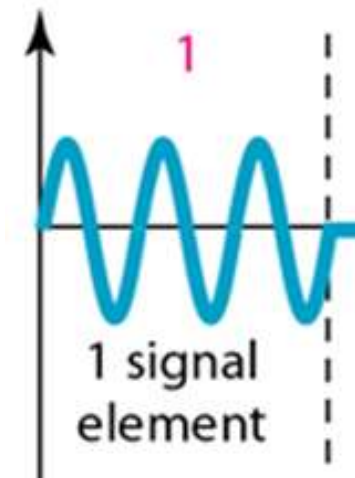


Types of Digital-to-Analog Conversion



Data and Signal Elements in Analog

- Data element: bit
- Signal element: the smallest unit of signal
- Signal element in digital signal
 - The smallest unit of signal which has a constant level
- Signal element in analog signal
 - The smallest unit of signal that represents one or more bits
 - In analog, always bit rate $\geq$ baud rate



Data Element and Signal Element

- N: Data transmission rate (bit rate)
- S: Signal transmission rate (baud rate)

$$S = N \times \frac{1}{r}$$

Example

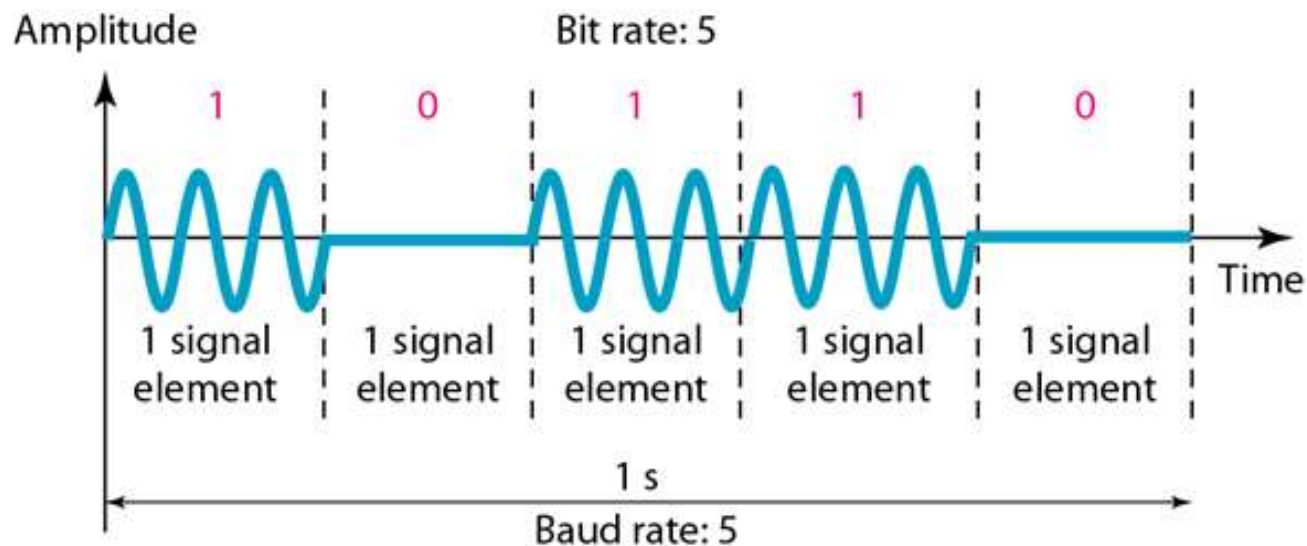
- An analog signal carries 4 bits of data per signal element. If baud rate is 1000 baud, what is the data transmission rate?

Example

- An analog signal transmits at 1000 baud and its bit rate is 8000 bps. How many bits are carried in a single signal element?
- How many different signal elements must be used?

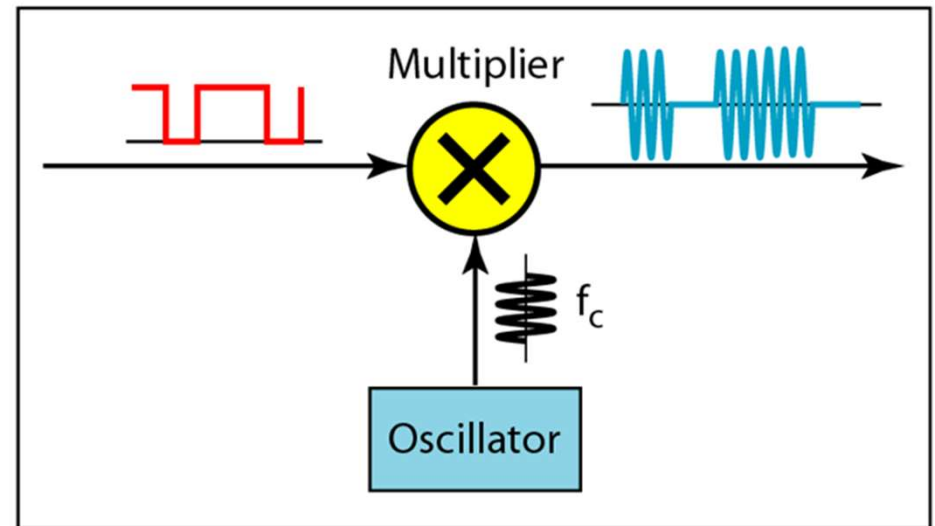
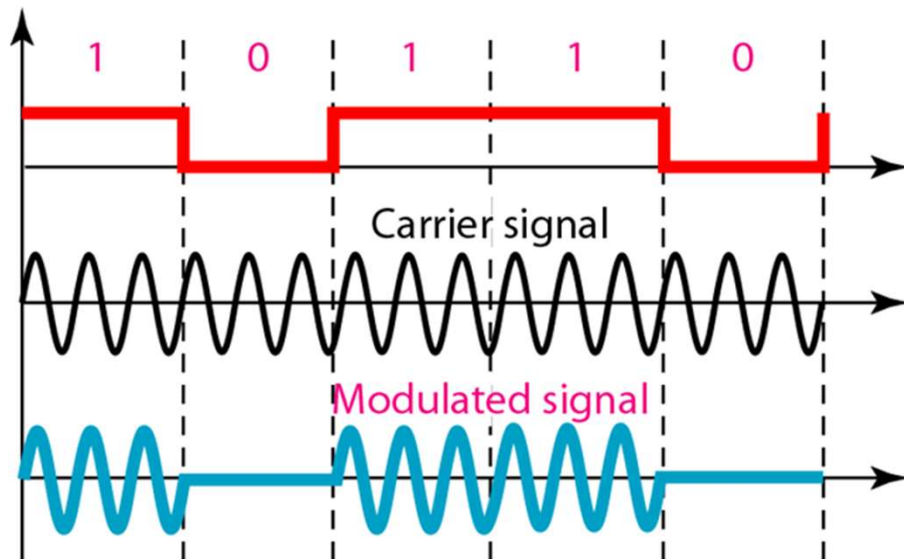
Amplitude Shift Keying (ASK)

- Change amplitude of the carrier signal to encode data
- Carrier signal: the analog signal that carries the data
- Carrier frequency: frequency of the carrier signal (typically, the central frequency)



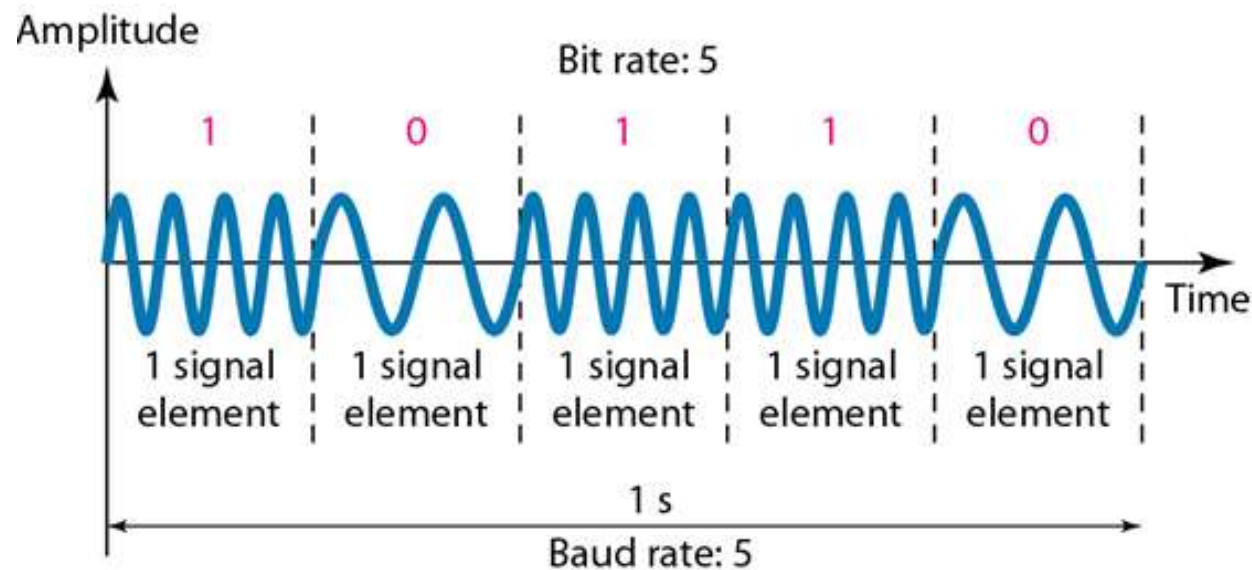
ASK

- Signal generation



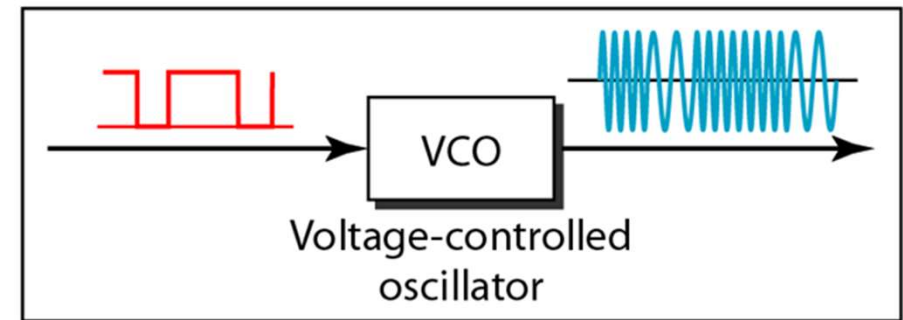
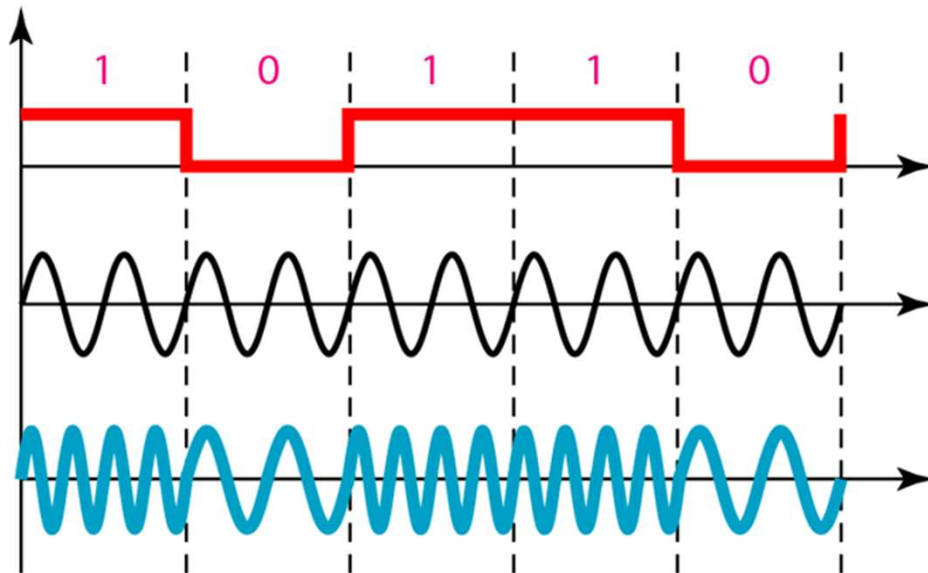
Frequency Shift Keying (FSK)

- Change frequency of the carrier signal to encode data



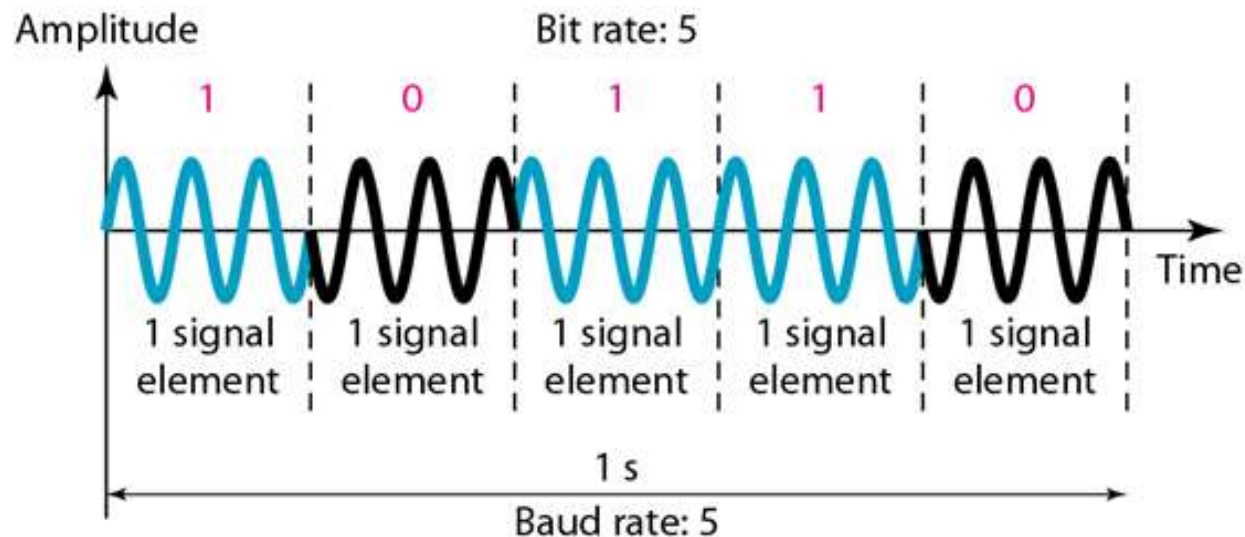
FSK

- Signal generation



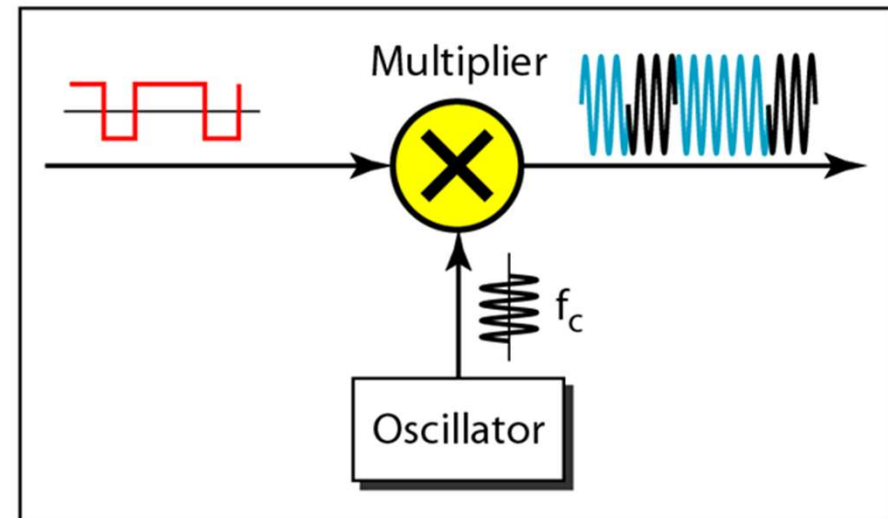
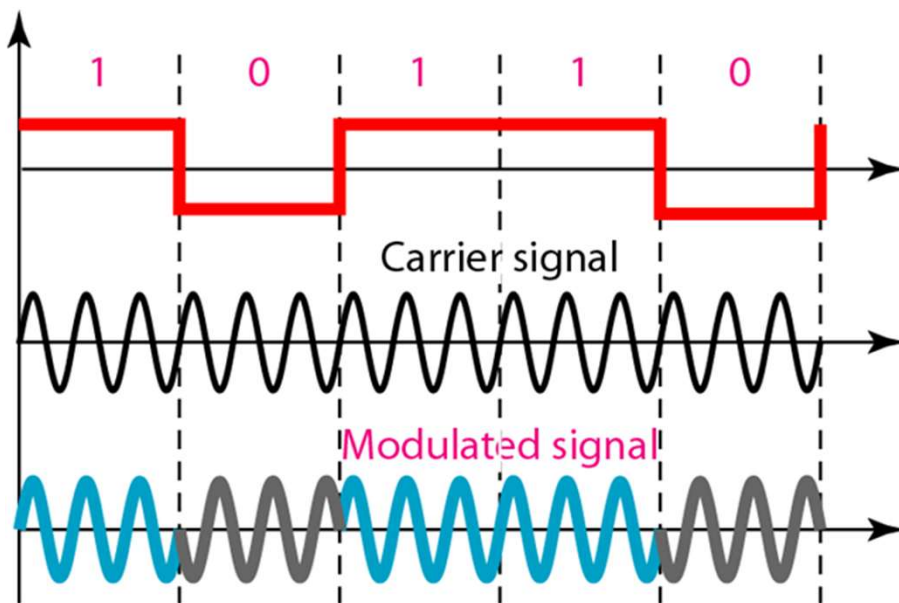
Phase Shift Keying (PSK)

- Change phase of the carrier signal to encode data



PSK

- Signal generation
 - ASK uses unipolar input (0, 1), PSK uses bipolar input (+1, -1) - same multiplier, different results

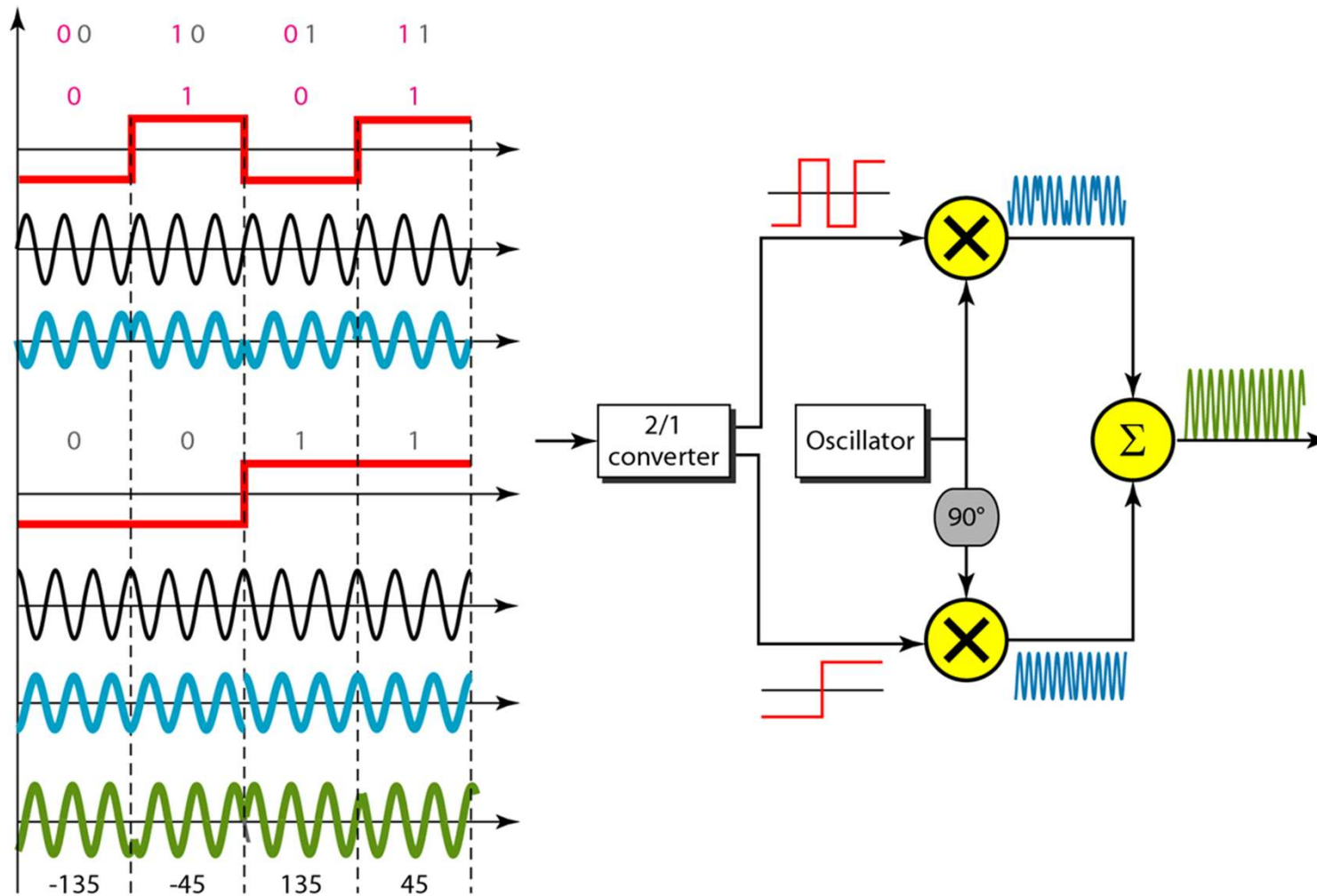


BPSK vs. QPSK

- BPSK (Binary PSK): use 2 different phases to encode 0 and 1
 - 0° , 180°
- QPSK (Quadrature PSK): use 4 different phases to encode 00, 01, 10, 11 (2 bits per signal element)
 - 45° , 135° , -45° , -135°

QPSK

- Signal generation

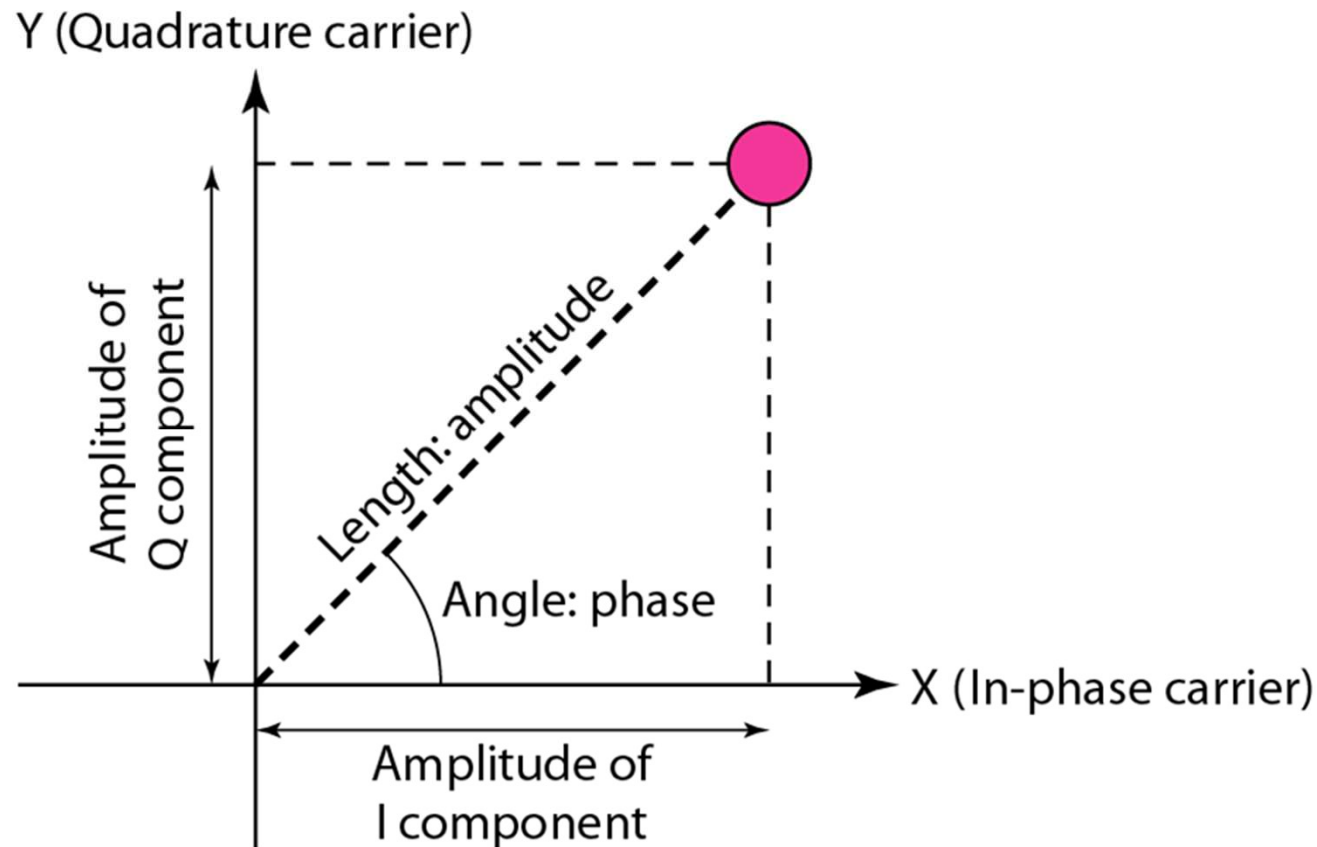


Example

- Data was transmitted at 12Mbps using QPSK modulation. What is the baud rate?

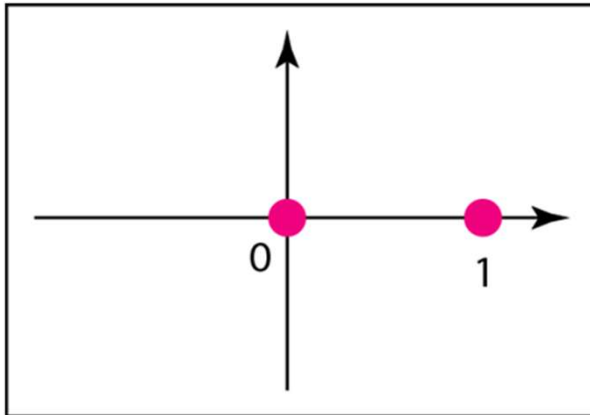
Constellation Diagram

- A diagram to represent amplitude and phase of a signal

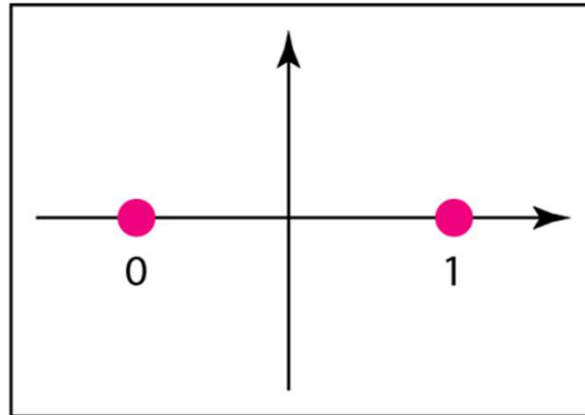


Constellation Diagram

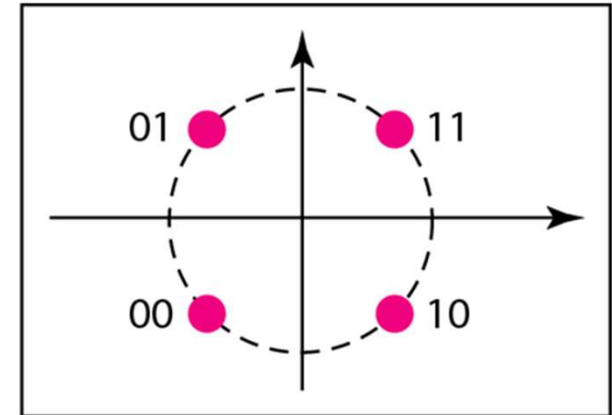
- ASK, BPSK, QPSK



a. ASK (OOK)



b. BPSK



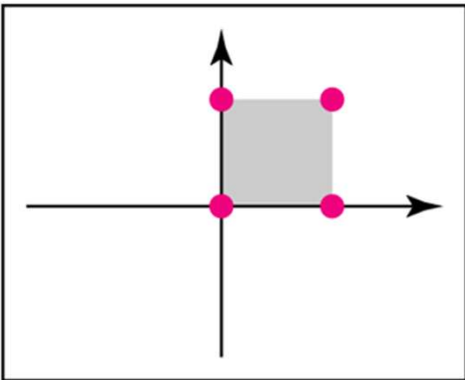
c. QPSK

Quadrature Amplitude Modulation

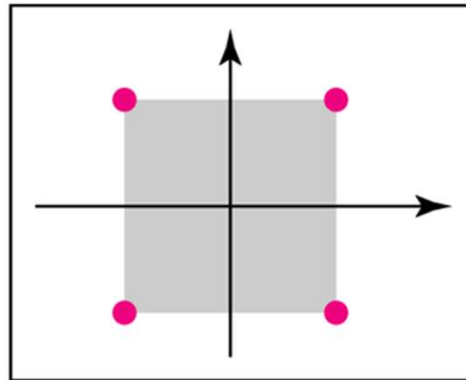
- QAM combines amplitude and phase variation on the carrier signal
- QPSK: I and Q each use only 2 levels (+1, -1)
- QAM: I and Q each use multiple levels
 - e.g., 16-QAM: I, Q $\in \{-3, -1, +1, +3\} \rightarrow 4 \times 4 = 16$ points
 - e.g., 64-QAM: I, Q $\in \{-7, -5, -3, -1, +1, +3, +5, +7\} \rightarrow 64$ points
- n-QAM: n points on constellation $\rightarrow \log_2 n$ bits per symbol
- More points $\rightarrow$ higher bit rate at the same baud rate, but closer spacing $\rightarrow$ more susceptible to noise

QAM

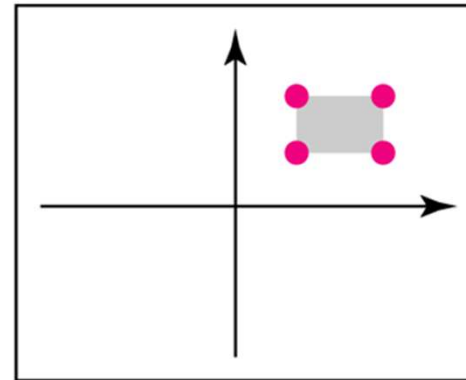
- n-QAM: use **n** points on the constellation diagram
 - use n signal elements to encode $\log_2 n$ bits



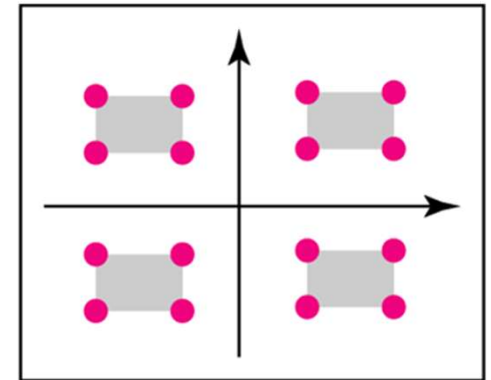
a. 4-QAM



b. 4-QAM



c. 4-QAM



d. 16-QAM

QAM: From Modems to Modern Wireless

- Higher QAM order = more bits/symbol in the same bandwidth
- 4096-QAM: 4096 points on constellation → requires very high SNR (~40dB)
- Trade off: more points = more bits, but less noise margin

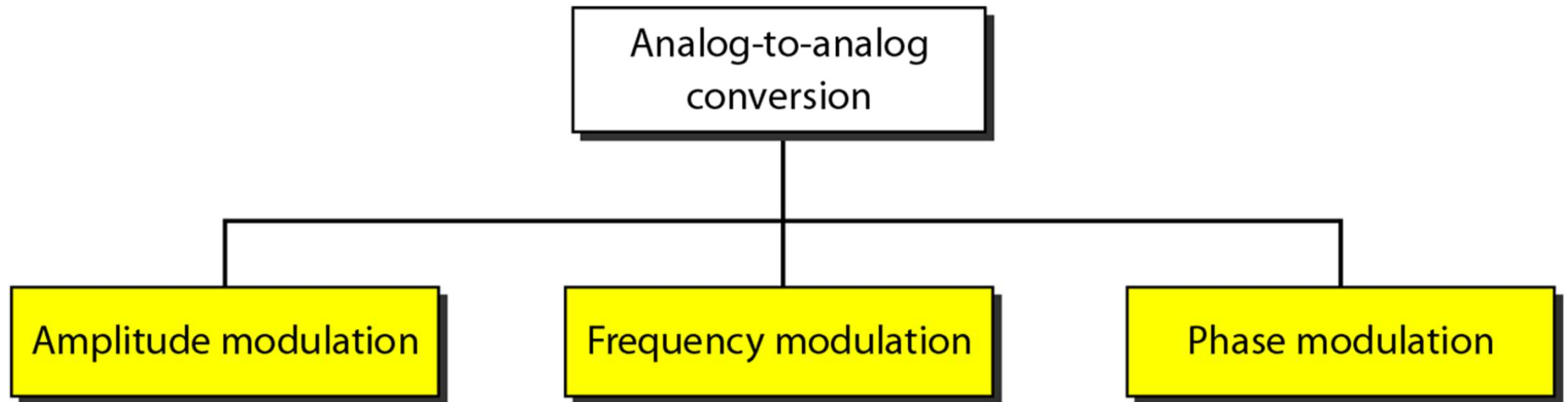
Technology	QAM Order	Bits/Symbol
Dial-up model V.22bis (1984)	16-QAM	4
Wi-Fi 5 (2013)	256-QAM	8
Wi-Fi 6 (2019)	1024-QAM	10
Wi-Fi 7 (2024)	4096-QAM	12
DOCSIS 4.0	4096-QAM	12

Analog-to-Analog Conversion

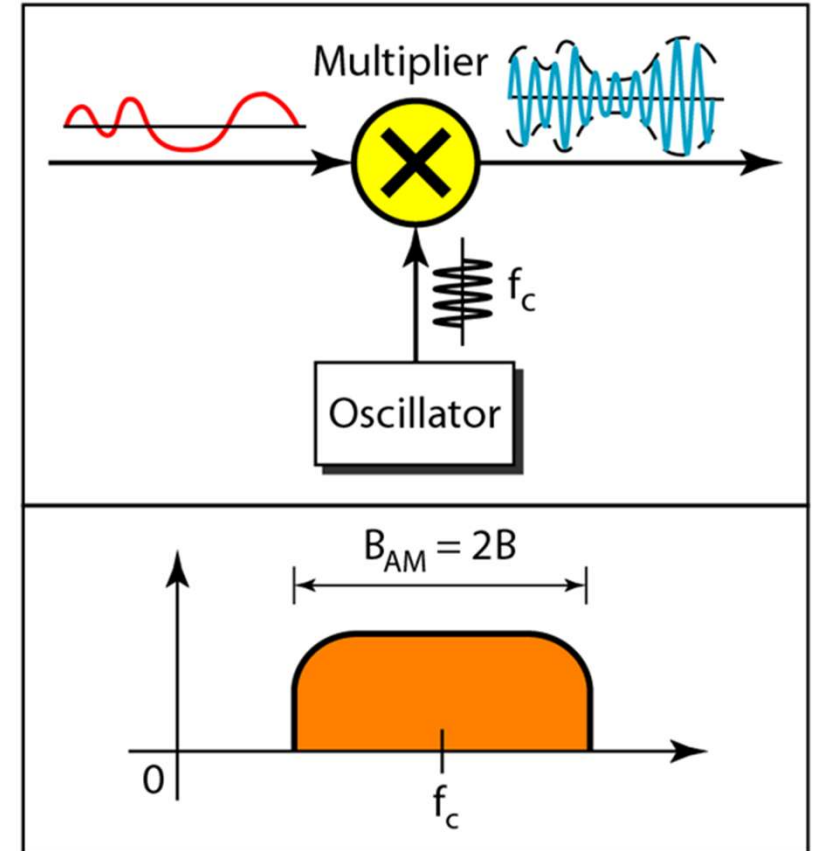
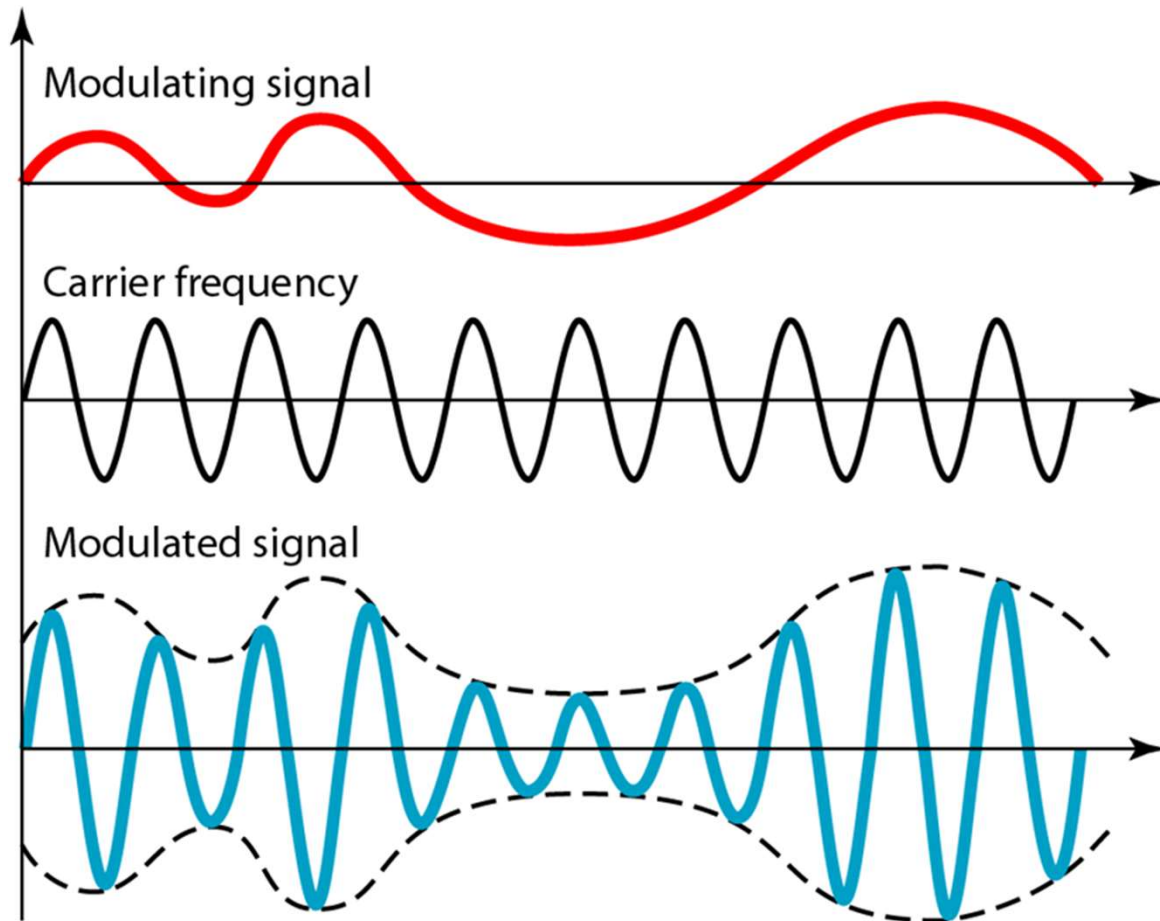
Analog-to-Analog Conversion

- Why convert?
 - To send analog data using carrier frequency
- Example: FM radio
 - Audio has a frequency range of 20-20,000 Hz
 - But radio channel is on a different frequency band
 - 88 – 108 MHz
 - Thus, we need analog-to-analog conversion

Analog-to-Analog Conversion

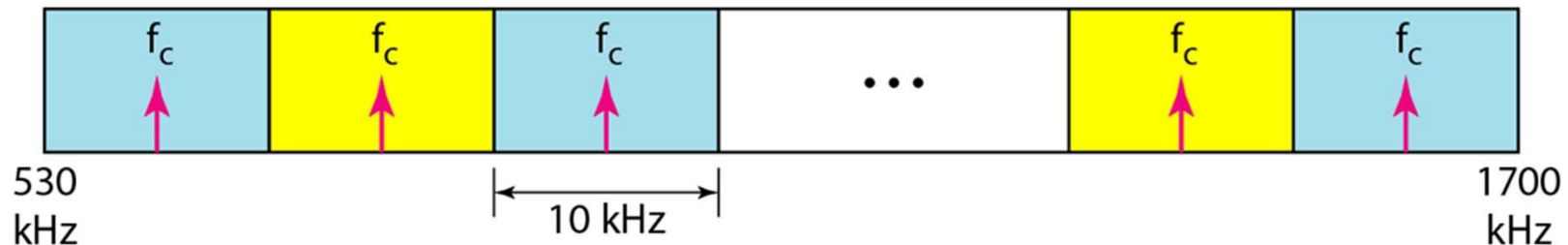


Amplitude Modulation

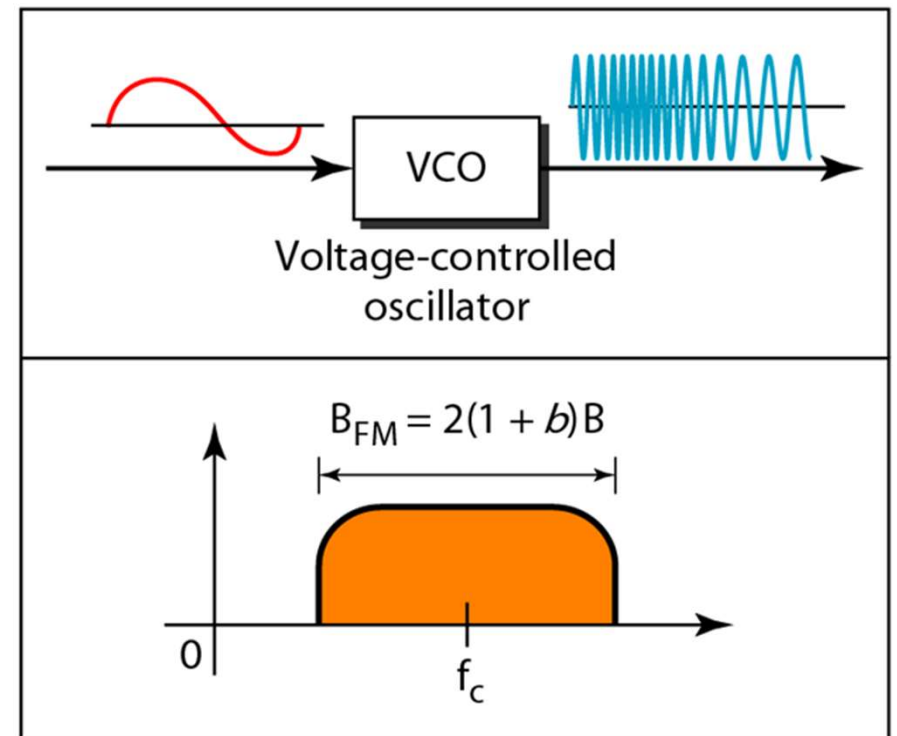
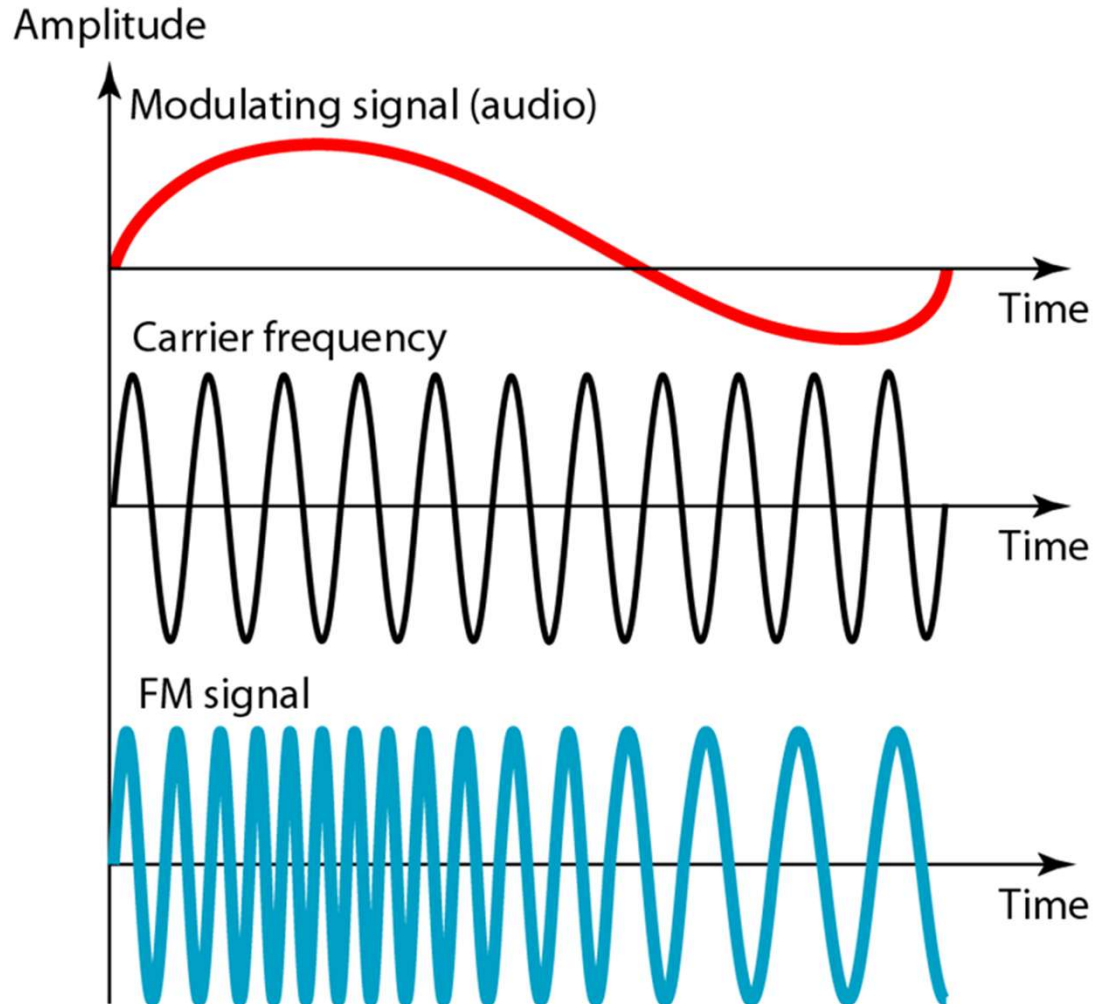


Amplitude Modulation

- AM Radio
 - 10kHz per channel
 - Carrier frequency: 530kHz ~ 1700kHz
 - Frequency gap between channels: at least 10kHz

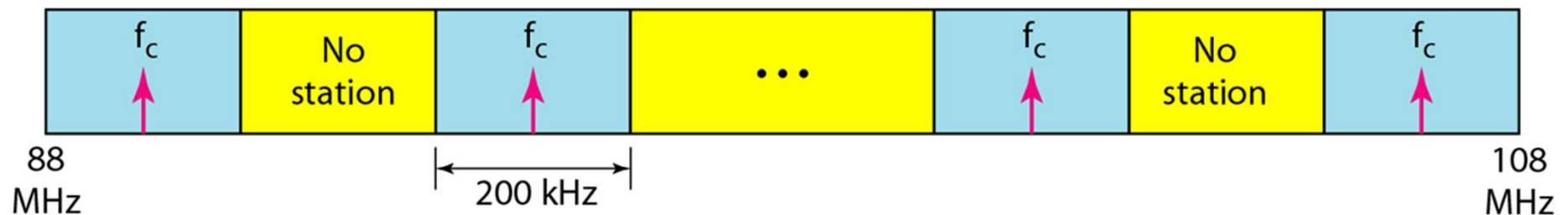


Frequency Modulation

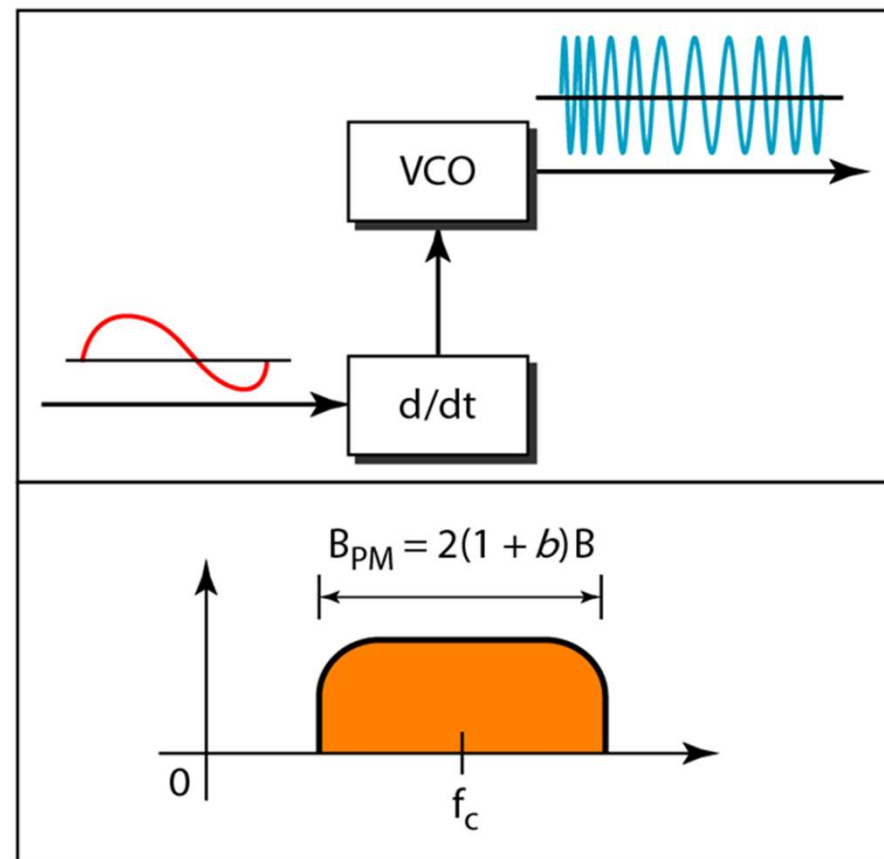
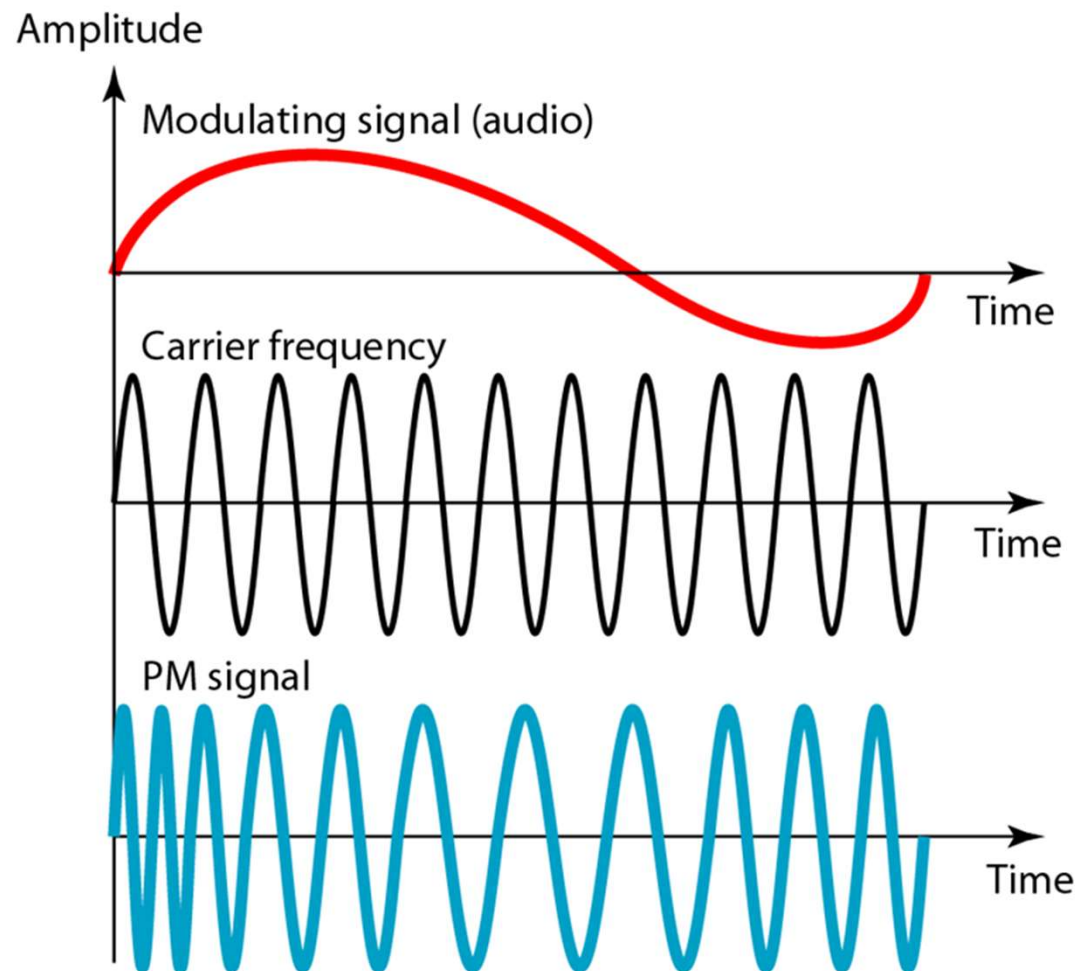


Frequency Modulation

- FM Radio
 - 200kHz per channel
 - Carrier frequency: 88MHz – 108MHz
 - Frequency gap between channels: at least 200kHz
 - In practice, channels are 400kHz apart (regulation to avoid interference)



Phase Modulation



End of Chapter
